

Materials Science and Engineering Concentration



What is Materials Science and Engineering?

- The study of materials:
 - Their structures, properties, behavior, mechanisms, synthesis, processing, etc.
 - Their applications.
- High-performance materials form the basis of our most advanced technologies:
 - Batteries (electrodes, electrolytes)
 - Medical implants (bio-compatible alloys, polymers and ceramics)
 - Propulsion systems (refractory alloys and ceramics)
 - Solar cells, electronic devices, displays (semiconductors)
- A fundamentally **multidisciplinary** field of study:
 - Our graduates work in the aerospace, transportation, electronic, biotech, ... industries.
- **Enjoy the benefits of an exceptionally high faculty-to-student ratio!**

The Materials Science Approach

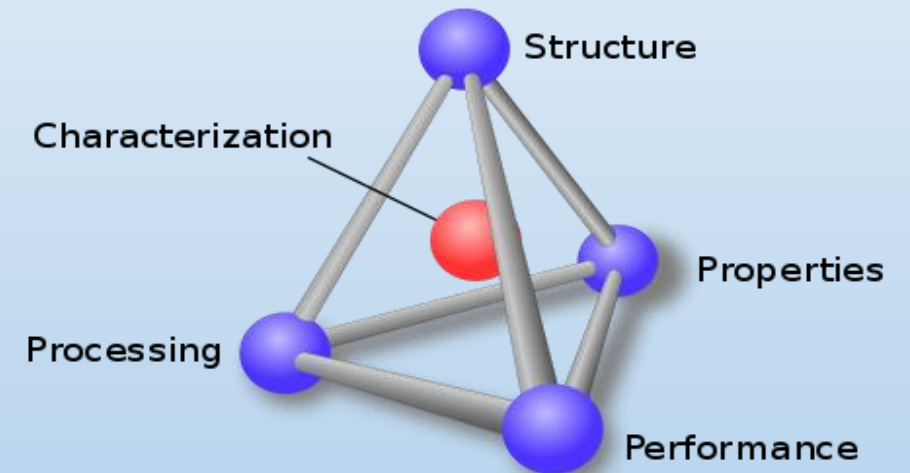
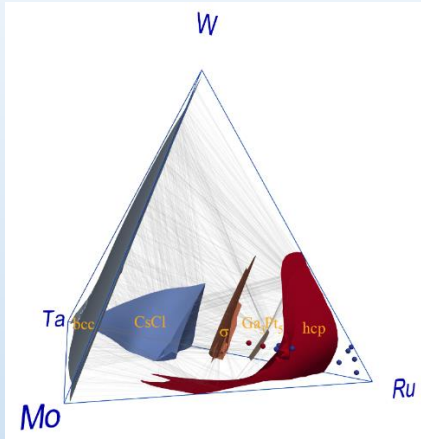
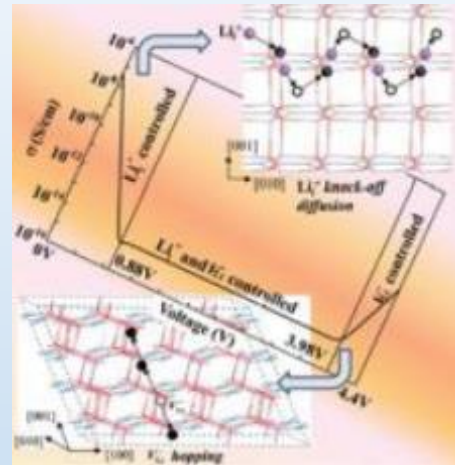


Image from http://mscunila.blogspot.com/2009_04_01_archive.html

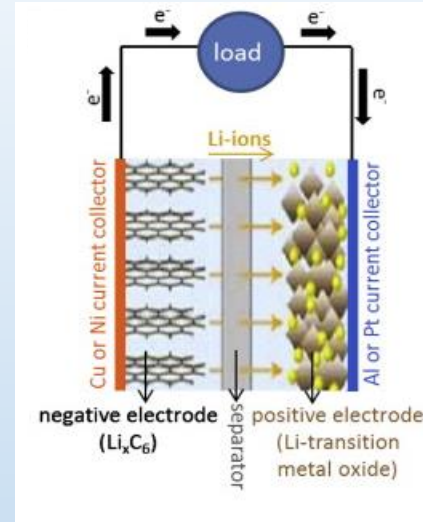
Materials science and engineering, in pictures



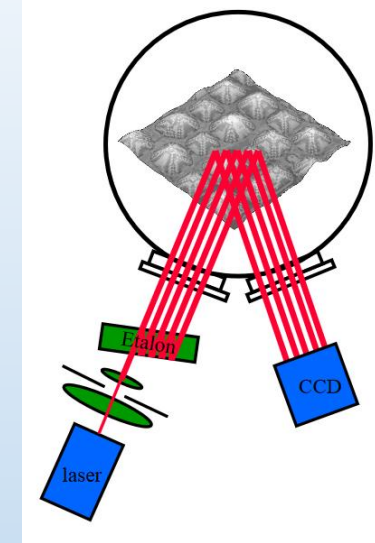
High-throughput Phase diagram modeling (van de Walle group)



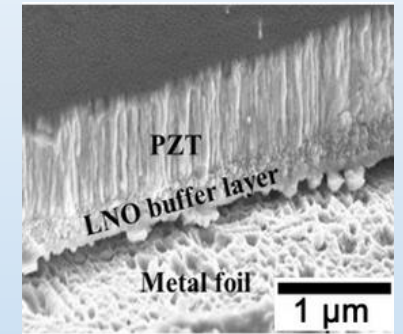
Electron and ion transport in complex materials (Qi group)



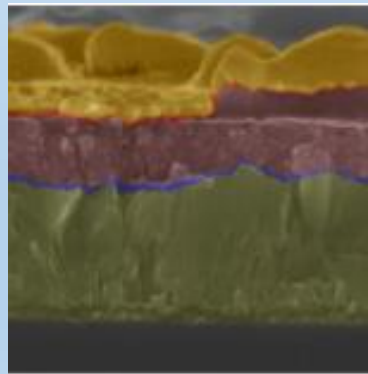
Stress in batteries (Sheldon group)



Surface Stress Measurements (Chason group)



Piezoelectrics (Kington group)



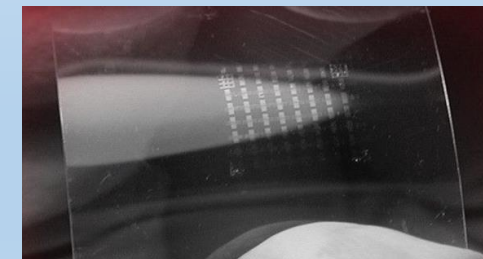
Hybrid Perovskite Solar cell (Padture group)



Cryogenic fuel tanks (Kumar group)



Advanced polymers For batteries (Palmore group)



Transparent conducting oxides (Paine group)

A Sample of Job Market Outcomes

- Recent cohort:
 - Engineering: Lockheed Martin (aerospace); BioLite, Kairos Power (power generation).
 - Graduate Studies: MIT (PhD, Materials), Brown (ScM, Electrical engineering).
 - Management: Deloitte (Business analysis).
- Examples in recent history (2013-2017):
 - Engineering: General Dynamics (transportation), Rocketdyne (aerospace), BD Medical (medical).
 - Management: McKinsey & Company, DunesPoint Capital (VP), J. P. Morgan (fund manager, VP).
- Data from the Bureau of Labor Statistics (BLS):

Materials Engineers

Percentile	10%	25%	50% (Median)	75%	90%
Hourly Wage	\$25.27	\$33.85	\$47.98	\$62.24	\$76.91
Annual Wage	\$52,560	\$70,400	\$99,800	\$129,470	\$159,970

All Engineers

Percentile	10%	25%	50% (Median)	75%	90%
Hourly Wage	\$21.77	\$29.28	\$39.15	\$52.87	\$68.56
Annual Wage	\$45,280	\$60,910	\$81,440	\$109,970	\$142,610

Required Courses

- Interdisciplinary core courses for the Sc.B. (first 2 years).
- 3 upper level engineering courses:
 - Thermodynamics
 - Kinetics
 - Mechanical properties of materials
- 3 out of 5 topical materials courses:
 - Electronic materials
 - Soft and polymeric materials
 - Ceramics and composites
 - Biomaterials
 - Metallic materials
- 1 advanced science course, e.g.:
 - Physics of Matter, Organic or Quantum Chemistry.
- Capstone design project.
- Four approved humanities courses

The Faculty



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Concentration courses (in more depth):

2. Upper-Level Materials Engineering Curriculum

ENGN 1410	Physical Chemistry of Solids (thermodynamics)	1
ENGN 1420	Kinetics Processes in Materials Science and Engineering	1
ENGN 1440	Mechanical Properties of Materials	1
PHYS 0790	Physics of Matter	1
or CHEM 0350	Organic Chemistry I	
or CHEM 1140	Physical Chemistry: Quantum Chemistry	
Three of the following: ²		3
ENGN 1450	Properties and Processing of Electronic Materials	
ENGN 1470	Composite Materials	
ENGN 1475	Soft Materials	
ENGN 1480	Metallic Materials	
ENGN 1490	Biomaterials	

3. Capstone Design

ENGN 1000	Projects in Engineering Design I
or ENGN 1001	Projects in Engineering Design II
or ENGN 1930L	Biomedical Engineering Design and Innovation

4 humanities and social sciences classes

Interdisciplinary core courses for the Sc.B. (first 2 years)

1. Core Courses:

ENGN 0030	Introduction to Engineering
or ENGN 0031	Honors Introduction to Engineering
ENGN 0040	Dynamics and Vibrations
ENGN 0410	Materials Science
ENGN 0510	Electricity and Magnetism
ENGN 0520	Electrical Circuits and Signals
ENGN 0720	Thermodynamics
ENGN 0310	Mechanics of Solids and Structures
or ENGN 0810	Fluid Mechanics
CHEM 0330	Equilibrium, Rate, and Structure
MATH 0190	Advanced Placement Calculus (Physics/Engineering)
or MATH 0170	Advanced Placement Calculus
MATH 0200	Intermediate Calculus (Physics/Engineering)
or MATH 0180	Intermediate Calculus
or MATH 0350	Honors Calculus
APMA 0330	Methods of Applied Mathematics I, II
or APMA 0350	Applied Ordinary Differential Equations

APMA 0340	Methods of Applied Mathematics I, II
or APMA 0360	Applied Partial Differential Equations I
or MATH 0520	Linear Algebra
or APMA 1210	Operations Research: Deterministic Models
or APMA 1650	Statistical Inference I
CHEM 0350	Organic Chemistry
or CSCI 0040	Introduction to Scientific Computing and Problem Solving
or CSCI 0111	Computing Foundations: Data
or CSCI 0150	Introduction to Object-Oriented Programming and Computer Science
or CSCI 0170	Computer Science: An Integrated Introduction
or CSCI 0190	Accelerated Introduction to Computer Science
or ENGN 1230	Instrumentation Design
or ENGN 1740	Computer Aided Visualization and Design
or ENGN 1750	Advanced Mechanics of Solids
or APMA 0160	Introduction to Scientific Computing

Useful links:

- These slides: <https://alum.mit.edu/www/avdw/concfair.pdf>
- Materials Concentration advisor: Axel van de Walle avdw@brown.edu
- <https://engineering.brown.edu/undergraduate/concentrations/materials-science-and-engineering>
- <https://bulletin.brown.edu/engineering/#undergraduate>

